

Climate Change & Antimicrobial Resistance

Western Pacific Region

Mixed-methods systematic analysis · Jan 2000 – Mar 2025

BACKGROUND

Two escalating threats
Climate change · AMR

↓ Climate
Rising temps

🦠 AMR
Resistance

Intersection mechanisms

Direct pathway

↑ temp → faster bacterial
growth & mutation rates
↑ resistance gene spread

Indirect pathway

Extreme weather events
→ healthcare disruptions
→ antibiotic misuse
→ AMR amplification

Western Pacific Region

Unique climatic
& socioeconomic
vulnerabilities

AMR mortality burden

GRAM project data used
to visualise trends

METHODS

3-stage mixed-methods
Systematic analysis

1

Narrative review

AMR landscape mapping
Climate-driven mechanisms
Western Pacific focus

2

Systematic review

PubMed + Google Scholar
Jan 2000 – Mar 2025
18 studies · SWiM synthesis
Quantitative regional studies

3

Empirical analysis

Longitudinal panel dataset
AMR mortality trends (GRAM)
Regression: temp & socioec.
factors vs AMR deaths
Climatic + socioec. covariates

Evidence synthesis

Qualitative + quantitative

Key pathogens studied

CRAB · CRPA
Carbapenem-resistant species

FINDINGS & IMPLICATIONS

Key quantitative findings

+1 °C temperature → ↑ AMR deaths

CRAB mortality

Carbapenem-resistant
A. baumannii

$$\beta = 0.652^{**}$$

95% CI 0.579–0.724

CRPA mortality

Carbapenem-resistant
P. aeruginosa

$$\beta = 0.422^{***}$$

95% CI 0.304–0.541

Literature findings

18 studies synthesised
↑ temp → ↑ clinical resistance
↑ ARG environmental spread

Socioeconomic factors

Heterogeneous effects
across subgroups

Environmental AMR

ARG dissemination enhanced
by higher temperatures

Interpretation & recommendations

Climatic conditions and socioeconomic vulnerabilities jointly shape AMR risk in the Western Pacific Region

Climate-resilient health systems

Strengthen infrastructure to withstand
extreme weather disruptions
and maintain antimicrobial stewardship

Multisectoral AMR governance

Improve cross-sector coordination
integrating climate and health
policy agendas regionally

Integrated surveillance networks

Establish AMR–climate surveillance
systems to monitor and respond
to intersection risks

⚠️ Projected threat: increasing extreme weather events → further strained health systems + worsened antibiotic misuse
Evidence gap identified · Regional priority: integrated response to climate–AMR intersection

Key terms:

CRAB

CRPA

AMR

ARGs

GRAM

SWiM

*** $p < 0.001$ · CRAB: carbapenem-resistant *A. baumannii* · CRPA: carbapenem-resistant *P. aeruginosa* · ARG: antibiotic resistance gene
SWiM: Synthesis Without Meta-Analysis · GRAM: Global Research on Antimicrobial Resistance